

E8: characteristic-two Kani preflight

C37 inert conductor step | source audit and exact arithmetic

Question. Does Galbraith's construction in Climbing and Descending Tall Isogeny Volcanos (ePrint 2024/924, Sections 3-4) apply unchanged to the degree-73 conductor step over characteristic two? The input is the surface curve $y^2 + xy = x^3 + 1$ over $F_{2^{37}}$, registry ID icv1-f2m37-tm534059-32aad96b, trace -534059.

The exact parity gate

The paper's Section 3 requires the Kani degree M and both input isogeny degrees N and m to be coprime to the characteristic. Section 4 sets $M = N + m$. For odd N , these two remaining requirements cannot hold together in characteristic two:

Choose odd M

Then $m = M - N$ is even. The auxiliary isogeny violates the stated coprimality condition.

Choose odd m

Then $M = N + m$ is even. Full rank-two étale $E[M]$ is unavailable on an ordinary curve in characteristic two.

Both branches leave the cited proof's hypotheses. This does not rule out a different characteristic-two construction using a non-étale kernel or another auxiliary map.

Exact C37 calculation

A native Rust check computes $\pi = \tau^{37}$ with $\tau^2 + \tau + 2 = 0$. For $M = 81 = 9 \times 9$, $m = M - 73 = 8 = 2^2 + 2^2$. The candidate's auxiliary degree is even. The modular Frobenius values are:

mod 9: $\pi = -\tau$, order 24. mod 81: $\pi = 27 - \tau$, order 216.

Full $K_0[9]$ is rational over $F_{2^{(37 \times 24)}}$. This arithmetic neither constructs a degree-73 isogeny nor transfers a point.

Source audit and remaining obligation

Correction to the earlier E8 note

The first pass treated full rational $E[M]$ over a single compositum as a required input. Section 4 of ePrint 2024/924 instead works prime by prime with Frobenius eigenbases and avoids materializing that compositum in its efficient implementation. Torsion-field costs require an audit of any actual characteristic-two adaptation; the blanket full-M objection was too strong.

Evidence

The protocol and native tool were committed before the retained replay (final source commit 8e7d9a6148df). Three exact-arithmetic tests passed with Rust 1.93.1. The retained JSON receipt is byte-identical to an earlier development calculation; SHA-256: 6118d92f89aef25c0310e83f37405f56f414ed5b394578898ccdbe25f3c83a5e. Source, protocol, JSON and the full report are in [research/isogeny_conductor_gap_20261007/](https://research.isogeny_conductor_gap_20261007/).

Remaining work

To advance E8 to the C37 prototype, establish a characteristic-two theorem or algorithm that handles the auxiliary map and kernel, construct an explicit isogeny between registered endpoints, verify its action on the named subgroup, and count complete setup and transfer operations. Only then can the $n=131$ cost model be evaluated. There is no operation-cost or discrete-logarithm result from this preflight.

Primary source

Steven D. Galbraith, Climbing and Descending Tall Isogeny Volcanos, extended ePrint 2024/924, Sections 3-4 and Theorem 1. <https://eprint.iacr.org/2024/924>